

Chapter 130 — Management Strategies for Non-Muscle-Invasive Bladder Cancer (Ta, T1 and CIS)

INTRODUCTION

- **Non-muscle-invasive bladder cancer (NMIBC)** = malignant urothelial tumours that **have not invaded the detrusor**. It spans the relatively **benign course of low-grade papillary tumours** and the **aggressive course of high-grade disease**: urothelial carcinoma in situ (CIS) and high-grade Ta and T1.
- **Epidemiology (United States)**: over 80,000 new bladder cancers annually; **70–80% are non-muscle-invasive at presentation** — of these **60–70% Ta, 20–30% T1, about 10% CIS**. Incidence has declined over two decades, with trends varying by race and ethnicity.

Presentation

- **Commonest presentation: painless haematuria**. Bladder cancer is **far more frequent with gross than microscopic haematuria**; microscopic haematuria is defined as **≥3 red blood cells per high-power field**.
- A large multicentre prospective study confirmed **visible haematuria carries a several-fold higher cancer yield than non-visible**; prevalence with non-visible haematuria is low (**0.3–4.7%**).
- **Females experience delays in haematuria evaluation**, which worsens bladder cancer outcomes.
- **Irritative voiding symptoms are associated with CIS**: about **80% of CIS patients have irritative symptoms**, and their presence **doubles the risk of CIS** in haematuria patients.
- Hence **cystoscopy and upper tract imaging are indicated for haematuria and/or unexplained irritative symptoms**.

Natural History

- **Survival is often favourable, but the disease is defined by frequent recurrence** requiring regular cystoscopy; managed by **transurethral surgery, often followed by intravesical immunotherapy or chemotherapy**.
- **Recurrence risk about 40–60%**, influenced by patient and disease factors and by **quality metrics — completeness of resection and appropriate intravesical therapy**.
- **Progression is the key concern**: from about **6% in low-grade Ta to 17% in high-grade T1**. Hence risk stratification is paramount.

Concept — NMIBC is two diseases under one label: a **recurrence problem (low grade)** and a **progression problem (high grade)**; every management decision in the chapter follows from which of the two a patient has.

PATHOLOGY: GRADING, STAGING AND IMPLICATIONS FOR CLINICAL RISK STRATIFICATION

Pathological Staging

- **Staging follows the American Joint Committee on Cancer (AJCC) criteria**.
- **Clinical stage** = findings at **transurethral resection of bladder tumour (TURBT)** + imaging + **bimanual examination at TURBT**; it establishes whether **lamina propria and/or muscularis propria are present and involved**, and should assess **lymphovascular invasion (LVI)**.
- **Pathological stage** = partial or radical cystectomy specimen **plus pelvic lymph nodes**.

- **Three histological layers:** urothelium → **lamina propria** (suburothelial connective tissue) → **detrusor/muscularis propria**, which is **absent beneath the urothelium of diverticula**.
- **Tumour, node, metastasis (TNM) staging of NMIBC:**
 - **Ta** = non-invasive papillary tumour confined to urothelium, **regardless of grade**.
 - **T1** = invades **lamina propria** (subepithelial tissue).
 - **CIS (Tis)** = flat, high-grade, confined to urothelium.
- **T1 substaging:** extent of lamina propria invasion correlates with progression. Depth relative to the **muscularis mucosae** (discontinuous wispy smooth-muscle bands within the lamina propria) — **invasion deep to it carries worse prognosis** — but it **cannot be assessed in all TURBT specimens** (discontinuity, poor orientation).
- **T1a/T1b substaging has been proposed;** the **AJCC Staging Manual stresses assessing extent of lamina propria invasion**, but **no method is universally accepted**.

FIGURE 130.1 — STAGING OF BLADDER CANCER

- **CIS** = high-grade, **flat** malignancy confined to urothelium; **Ta** = papillary, confined to urothelium; **T1** = papillary invading lamina propria — it may **intertwine with muscularis mucosae fibres** but by **definition does not invade detrusor**; **T2** invades detrusor; **T3** reaches extravesical fat.

Pathological Grading

- **Current system:** World Health Organization/International Society of Urological Pathology (WHO/ISUP) 2015 (building on the 2004 revision). "**Urothelial carcinoma**" is preferred over "**transitional cell carcinoma**".
- Since 2004 grading is **two-tiered** — **low grade vs high grade** — replacing the **three-tiered WHO 1973 (Ash-Broder) system**.
- **Current categories:** hyperplasia (flat and papillary), reactive atypia, atypia of unknown significance, urothelial dysplasia, urothelial CIS, urothelial papilloma, **papillary urothelial neoplasm of low malignant potential (PUNLMP)**, non-muscle-invasive low-grade papillary urothelial carcinoma, non-muscle-invasive high-grade papillary urothelial carcinoma.
- **PUNLMP:** papillary tumour with **orderly arrangement, minimal architectural abnormality and minimal nuclear atypia**; distinct from both grades, **unlikely to progress, essentially benign**. The **WHO recommends a report note** that such patients are **at risk of recurrence**, occasionally as urothelial carcinoma (UC), so **follow-up is warranted**.
- **Papilloma:** truly benign; no progression risk.

TABLE 130.1 ESTIMATES OF DISEASE PROGRESSION IN NON-MUSCLE-INVASIVE BLADDER CANCER: WORLD HEALTH ORGANIZATION/INTERNATIONAL SOCIETY OF UROLOGIC PATHOLOGY CONSENSUS CLASSIFICATION

Tumour type	Relative frequency (%)	Progression (%)	Deaths (%)
<i>Non-invasive</i>			
Papilloma	10	0–1	0
PUNLMP	20	3	0–1
Papillary cancer, low grade (TaG1)	20	5–10	1–5
Papillary cancer, high grade (TaG3)	30	15–40	10–25
<i>Invasive</i>			
Papillary cancer (T1G3)	20	30–50	33
<i>Carcinoma in situ</i>			

Tumour type	Relative frequency (%)	Progression (%)	Deaths (%)
Primary	10	>50	—
Secondary	90	—	—

From Donat SM, Urol Clin North Am, 2003.

- **Purpose:** shows the **stepwise rise in progression and death with grade** — the jump is between **low-grade Ta and high-grade Ta**, not between Ta and T1.
- **Pattern:** high-grade Ta already carries **mortality comparable to invasive disease** (10–25% vs 33%).
- **Traps:** **most CIS is secondary** (accompanies papillary tumour); **primary CIS progresses in >50%**; papilloma and PUNLMP are not zero-risk for recurrence even though progression is negligible.

FIGURES 130.2–130.5 — HISTOLOGY (×40)

- **PUNLMP (130.2):** thickened urothelium, normal cells and nuclei.
- **Low-grade Ta (130.3):** relatively normal cells with **irregularity and some nuclear differentiation**.
- **CIS (130.4):** severe cellular irregularity and nuclear pleomorphism; no lamina propria invasion.
- **High-grade T1 (130.5):** nests of abnormal cells, mitotic figures, invasion of lamina propria.

Tumour Biology

- **Grade is among the most important prognostic factors.**

	Low-grade Ta	High-grade T1
Recurrence	50–70%	>80%
Progression	<5%	50% within 3 years

- Behaviour is **grade-dependent rather than stage-dependent**: high-grade tumours progress at similar frequency whether Ta or T1.
- **Controversy:** small (≤ 3 cm) high-grade Ta is stratified **intermediate risk**, yet **recurs and progresses like high-risk Ta** (> 3 cm, multifocal, high grade).
- **Prognosis also correlates with** tumour size, multiplicity, **papillary vs sessile configuration**, LVI and **status of the remaining urothelium**.
- **Low-grade pTa may be almost benign and a separate disease entity** from high-grade tumours (different genetic imprints) — yet **low- and high-grade lesions coexist**.
- **UC is a field-change disease** (tumours at different times and sites). **Rarely, low-grade patients develop high-grade tumours years later** → **long-term surveillance is appropriate**.
- **Molecular subtypes:** RNA-expression classification of NMIBC; the **UROMOL project** defined **four molecular classes** with distinct expression profiles and phenotypes. **Clinical utility currently limited**; potential basis for precision risk assessment.

Pathological Characteristics and Implications for Clinical Management and Risk Stratification

Ta

- **Usually low grade; recurrence common** (especially if multiple), **progression rare**. About **7% of Ta tumours are high grade** (range 3–18%).
- **The most important risk factor for progression is grade, not stage.**

Carcinoma In Situ

- **Not "premalignant":** CIS is a **flat, non-invasive UC, high grade by definition**, and a **precursor of invasive high-grade cancer**.
- **Histology:** severely dysplastic urothelium, disorderly, high-grade nuclear atypia; **denudation from loss of cohesion** can hamper interpretation.
- **Dysplasia/atypia reports cause confusion:** mild forms are benign, but **severe/high-grade dysplasia = CIS**.
- **Untreated CIS progresses to muscle invasion in 40–83%**, especially with **papillary tumour**.
- **"CIS alone" harbours invasion at cystectomy in up to 20%; CIS with presumed T1 markedly increases upstaging at cystectomy.**
- **CIS is the second most important prognostic factor after grade. Multicentricity is ominous.**
- **Irritative symptoms** have been linked to diffuse disease, invasion and poor prognosis — **no consensus**.
- **Node-positive disease with CIS only at radical cystectomy: up to 3%.**

T1

- **Usually papillary on a narrow stalk; nodular/sessile appearance suggests deeper invasion.**
- **Increased recurrence/progression risk with: deep lamina propria invasion (especially muscularis mucosae), LVI, pyuria, bladder neck involvement.**
- **Hydronephrosis often indicates muscle invasion.**

Understaging

- **Understaging is common in high-grade, apparently non-muscle-invasive disease, especially T1;** cystectomy frequently reveals more extensive disease than the transurethral resection (TUR) specimen suggested.
- **Restaging TURBT of high-grade T1 finds muscle invasion in up to half.** Hence a **thorough second resection** — particularly when muscle was absent from the first specimen.

TABLE 130.2 RISK OF UNDERSTAGING WHEN CYSTECTOMY IS PERFORMED FOR PRESUMED NON-MUSCLE-INVASIVE DISEASE

Series type	Risk of understaging
Single-institution series (United States, Europe, Turkey, Canada)	15–50% (most 27–43%)
Multi-institutional cohorts	46–50%
Range quoted in body text	34–62%

Data from nine series, 2001–2012 (Stein et al., 2001, to Turker et al., 2012).

- **Purpose:** quantifies how often cystectomy for "NMIBC" reveals worse disease — **roughly a third to a half**.
- **Trap:** the body-text range (34–62%) and the tabulated range (15–50%) differ because they draw on **different series and denominators**; the message is the same — **clinical T1 is unreliable**.

Risk Stratification

- **Risk stratification is the primary tool for choosing treatment, surveillance schedule and counselling.**
- **European Organisation for Research and Treatment of Cancer (EORTC) risk tables:** probability of recurrence and progression at 1 and 5 years.
 - **Recurrence factors:** prior recurrence rate, number of tumours, tumour size.
 - **Progression factors:** T stage, concurrent CIS, grade — **grade must be underscored**.

- **Club Urológico Español de Tratamiento Oncológico (CUETO)** model: similar, developed in **bacille Calmette-Guérin (BCG)**-treated patients.
- **EORTC limitations:** cohorts had little BCG maintenance, perioperative mitomycin C (MMC) or restaging TURBT → **overestimates recurrence and progression**. Updated EORTC tool uses BCG-maintenance cohorts but is **deficient in CIS** and needs validation.
- **American Urological Association/Society of Urologic Oncology (AUA/SUO) guideline (2016, updated):** novel **low/intermediate/high** schema, **validated** for recurrence and progression. **BlaTUR** — free application providing guideline-based, risk-adapted care.

TABLE 130.3 EUROPEAN ORGANISATION FOR RESEARCH AND TREATMENT OF CANCER (EORTC) AND CLUB UROLÓGICO ESPAÑOL DE TRATAMIENTO ONCOLÓGICO (CUETO) SCORING MODELS

Factor	EORTC	CUETO
Number of tumours	✓	✓
Prior recurrence rate	✓	✓
T stage	✓	✓
Concurrent CIS	✓	✓
Tumour grade (WHO 1973)	✓	✓
Tumour diameter	✓	—
Age	—	✓
Sex	—	✓

From Sylvester RJ et al., Eur Urol 2006 (EORTC); Fernandez-Gomez J et al., J Urol 2009 (CUETO).

- **Five shared factors; EORTC adds size; CUETO adds age and sex and drops size.**
- **Both use WHO 1973 grading** — a trap when applying them to WHO 2004/2016 reports.
- **CUETO was derived in BCG-treated patients**, so it predicts better than original EORTC in BCG-era practice.

TABLE 130.4 AMERICAN UROLOGICAL ASSOCIATION RISK STRATIFICATION AND THERAPEUTIC INTERVENTIONS

Risk	Criteria	Therapeutic intervention
Low	LG solitary Ta ≤3 cm; PUNLMP	Single postoperative instillation of chemotherapy
Intermediate	LG Ta recurrence within 1 year; solitary LG Ta >3 cm; LG Ta multifocal; LG T1; HG Ta ≤3 cm	6-week induction intravesical chemotherapy or BCG; maintenance to 1 year
High	HG T1; recurrent HG Ta; HG Ta >3 cm; HG Ta multifocal; any CIS; any BCG failure in HG; any variant histology; any LVI; any HG prostatic urethral involvement	6-week induction BCG; maintenance BCG to 3 years. BCG-unresponsive: timely radical cystectomy; systemic pembrolizumab; nadofaragene firadenovec; gemcitabine + docetaxel

Disease parameters: size, focality, grade, stage, LVI, prostatic urethral involvement, variant histology, response to BCG. Risk stratification is dynamic and must be reassessed throughout the disease course. LG, low grade; HG, high grade.

- **Traps:** LG T1 is intermediate, not high; HG Ta ≤3 cm is intermediate; any CIS, any LVI, any variant histology = high risk regardless of other features.
- **Treatment tracks risk:** single instillation → induction ± 1-year maintenance → BCG with 3-year maintenance → cystectomy or salvage.

KEY POINTS: PATHOLOGY AND RISK STRATIFICATION

- Malignant tumours are **low or high grade, regardless of invasion**.
- High- and low-grade cancers are **essentially separate diseases** (genetics, behaviour, management).
- **The most important risk factor for progression is grade.**
- **CIS is a precursor and risk factor** for progression, invasion and metastasis.
- **PUNLMP** = orderly arrangement, minimal architectural abnormality, minimal nuclear atypia.
- **Risk stratification is essential** for treatment, counselling and surveillance.

Concept — **Grade drives progression; number, size and prior recurrence drive recurrence.** *Stage and CIS matter chiefly because they signal high-grade biology and understaging — which is why every risk model is built on that split.*

ENDOSCOPIC SURGICAL MANAGEMENT

Procedures

Diagnosis and Upper Tract Evaluation

- Tumours usually found at office cystoscopy for haematuria → map location, number and morphology and review before anaesthetic TUR. If found **incidentally on cross-sectional imaging**, may proceed directly to TURBT.
- **Field change** → upper tract evaluation at diagnosis and, by risk, in surveillance: **computed tomography (CT) urography** commonly; **retrograde pyelography** during TURBT if not done; **ureteroscopy** reserved for suspicious findings.
- Upper tract tumours occur in <5% of bladder cancer patients. **Risk factors: trigonal tumours, concomitant CIS, high-risk disease.**

FIGURE 130.6 — CT UROGRAM

- **Filling defect along the right lateral bladder wall** on CT urography, proven **high-grade Ta** at resection.

Transurethral Resection of Bladder Tumour

- TURBT is both **diagnostic and therapeutic**; regional or general anaesthesia, **with or without paralysis**; careful positioning to avoid lower-limb nerve injury; specimens **systematically obtained and labelled**.
- **Bimanual examination before and after TURBT: fixation or a persistent palpable mass after resection suggests locally advanced disease** — though its added value with modern imaging is **limited and possibly misleading**.
- **Technique sequence**: complete inspection with **flexible cystoscope and/or 30° and 70° lenses** → **continuous-flow irrigation, bladder filled just enough to efface folds** (overdistension thins detrusor → perforation; outflow suction refines volume control) → **piecemeal resection debulking intraluminal tumour** → **transect base or stalk, lifting the tumour edge away from detrusor** → **separate deep-margin pass or cold-cup biopsy of the base** sent separately for muscle invasion → **cautery for haemostasis** → **Ellik evacuator for chips** → **final haemostasis check at minimal irrigation**.
- **Resection quality determines outcome**, especially for high-grade disease: **complete visual resection** is recommended, yet **recurrence varies widely by surgeon**.
- **Muscle in the specimen** is required for staging; registry data show **nearly half of diagnostic resections lacked muscle** (or its mention), which translated into **increased mortality**.

Monopolar	Bipolar
Sterile water irrigation	Saline irrigation → lower TUR syndrome risk in long resection or perforation
Standard	Systematic review: shorter stay, less bleeding, fewer obturator reflexes/perforations; randomised trial: no difference in perforation, bleeding or obturator reflex, but less thermal artefact

- **No clear superiority of either energy modality.**

Special Circumstances

Diverticular tumours.

- **Harder to see and resect; no detrusor backing** → high perforation risk and unreliable staging.
- **Invasion beyond diverticular lamina propria = perivesical fat = T3a by definition. Clinical understaging in over half of Ta/T1 cases at cystectomy.**
- **Low grade:** conservative — **cold-cup resection and fulguration. High grade:** sample base including perivesical fat; **strongly consider partial or radical cystectomy. No consensus** on management.

Tumour location.

- **Anterior wall and dome (capacious bladder):** minimal filling + suprapubic compression; **extra-long resectoscope**; rarely a temporary perineal urethrostomy (obese patients); **digital rectal or vaginal manipulation**; angled or flat loops for posterior wall/dome.

Ureteric orifice.

- **Pure cutting current and limited cautery** → minimal scarring; **resection of the orifice is permissible.**
- **Resecting the intramural ureter** may eradicate tumour but **risks reflux of malignant cells** (clinical significance unclear).
- **Post-resection intramural stenosis about 12%; predictors:** tumour >1.5 cm and T1. T1 in the bladder and CIS in the intraluminal distal ureter raise upper tract UC risk.
- **Routine stenting unnecessary** after careful unilateral orifice resection.
- **Aids:** efflux check with **intravenous indigo carmine or methylene blue**, or retrograde study; with **fluorescence cystoscopy** the urine jet appears green; **open-ended ureteric catheter** before resection of adjacent tumours. **Stricture rare (about 4%);** most hydronephrosis resolves or reflects invasion. **Postoperative renal ultrasonography** if orifice resected or coagulation used nearby.

Small tumours and thin bladders.

- **Small or narrow-stalk tumours: cold-cup forceps alone** — useful in **thin-walled bladders (elderly women)**; perforation is smaller than with the loop. **Bugbee electrode** for haemostasis.

Laser Resection

- **Holmium and thulium lasers** most used; **equivalent safety and efficacy**; en bloc resection or vaporisation.
- **En bloc resection:** loop electrode, **HybridKnife**, holmium, thulium or **potassium titanyl phosphate (KTP) laser**. **Excises tumour with underlying muscle intact** → **preserved architecture, less cautery artefact, more accurate staging**; best for **papillary tumours on a discrete stalk**.
- **Meta-analyses:** fewer perforations, less residual disease, more detrusor in the specimen, fewer obturator kicks — **but no difference in recurrence. Endorsed by the European Association of Urology (EAU) guidelines** as a viable strategy.

- **Disadvantages:** large tumours need **fragmentation** for extraction; disruption at the resection line; **equipment cost**; excessive penetration depth.

Repeat Transurethral Resection of Bladder Tumour

- **Second-look TURBT is a quality indicator.** Rationale: **incomplete resection** (volume, access, complications) drives **early recurrence**; **muscle in the specimen** (itself a quality indicator) is critical where **lamina propria invasion** is found or invasion suspected.
- **Residual tumour is common:** tumour at first follow-up cystoscopy in **up to 45%**; **residual tumour at the resection site in $\geq 40\%$** when re-resected within weeks; second TURBT **corrects staging errors in about half** of those with residual disease.
- **High-grade Ta:** residual tumour in **up to 50%**, **upstaging up to 15%** → AUA: **consider repeat TURBT within 6 weeks for multifocal and/or large high-grade Ta.**
- **High-grade T1:** worse findings or **upstaging to muscle invasion in 25–30%**; **up to about half if no muscle in the first specimen** → **repeat TURBT within 6 weeks recommended.**
- **Understaging in T1 is about twice as likely when muscle is absent.**
- **Benefits:** second resection **changes treatment in one-third**; observational data show **better survival and fewer recurrences**; randomised data show **lower recurrence and progression in T1. Omission of muscle (or of its mention) → increased mortality.**
- **Residual T1 at re-resection** → **progression up to 80%** → consider **timely cystectomy.**
- **Second opinion:** re-resect unless clear muscle invasion, especially if outside slides unavailable; **subspecialty genitourinary pathology review changes management in about one-third** (interobserver variability).
- **AUA summary:** repeat TURBT of the primary site, including muscularis propria, within 6 weeks for incomplete initial resection and all T1; consider for high-risk high-grade Ta. Greatest benefit when no muscle was present initially.

Repeat TURBT: recommended	Repeat TURBT: consider
Incomplete initial resection	High-risk HG Ta (multifocal and/or large)
All T1 (especially no muscle)	
No muscle in specimen (AUA/EAU)	

Bladder and Urethral Biopsies

- **Biopsy all suspicious areas; cold-cup biopsies avoid cautery artefact.**
- **Positive cytology + erythematous velvety mucosa** → biopsies may diagnose CIS.
- **Random biopsies of normal-appearing mucosa: low yield. Not indicated in low risk** (low-grade papillary, negative cytology); **consider with high-risk history or positive cytology** (or multiple tumours).
- **Prostatic urethra:** risk low in low-risk disease, increased with bladder neck/trigone tumours or CIS.

Office-Based Endoscopic Management

- **Small (≤ 1 cm) low-grade recurrences** → **office thermal or laser ablation**, avoiding anaesthesia.
- **Analgesia:** 1–2% lidocaine with bicarbonate instilled, **15–30 minutes' dwell**; fulguration of 1–5 mm tumours is often tolerable without it.
- **Randomised non-inferiority trial:** office laser photocoagulation under photodynamic guidance was **non-inferior to TURBT** for recurrence-free survival, with **better quality of life.**

- **Prerequisite:** no tissue is obtained, so **prior tissue diagnosis and negative cytology** are **mandatory** to confirm low grade.
- **Active surveillance of small low-grade tumours** until significant growth is safe in selected patients. **Bladder Cancer Italian Active Surveillance (BIAS)** study: about **half remain treatment-free at 2 years**; **failure predicted by multiple prior TURBTs and multiple tumours**. Avoids anaesthesia, improves quality of life, limits financial toxicity.

Enhanced Cystoscopic Techniques: Fluorescence Cystoscopy, Narrow-Band Imaging, Optical Coherence Tomography and Confocal Laser Endomicroscopy

- **White-light cystoscopy (WLC)** is the **gold standard**, but subjective: a substantial proportion of biopsies of suspicious areas are negative, and **imperfect sensitivity partly explains early recurrence** after "complete" resection.
- **Enhancements:** fluorescence cystoscopy, narrow-band imaging (NBI), optical coherence tomography (OCT), confocal laser endomicroscopy.

Fluorescence (Blue-Light) Cystoscopy

- **Mechanism:** photoactive **porphyrins** accumulate in neoplastic tissue and **fluoresce red** under blue light.
- **5-aminolevulinic acid (5-ALA)** caused **systemic photosensitisation** → replaced by **hexaminolevulinate (HAL)**, which avoids it. Also termed **photodynamic diagnosis (PDD)**.
- **Detects small papillary tumours and almost one-third more CIS** than WLC; **greatest gain is for CIS**.
- **Recurrence:** earlier prospective data showed **reduced recurrence for several years**; a meta-analysis supports reduction **up to 1 year and beyond, inconsistently**. However, a **multicentre pragmatic randomised trial found no reduction in 3-year recurrence and higher cost**, mainly in intermediate risk. **Progression benefit unproven** (underpowered).
- **Also assesses completeness of TUR.** Office flexible blue-light cystoscopy detects **about 20% more recurrent papillary tumours and about one-third more CIS**.

FIGURES 130.7–130.8 — BLUE-LIGHT CYSTOSCOPY

- **Normal-appearing mucosa on white light shows red HAL fluorescence** under blue light, proven NMIBC — with rigid (130.7) and **flexible** (130.8) cystoscopes.

Narrow-Band Imaging

- **Mechanism:** two haemoglobin-absorbed wavelengths — **415 nm penetrates superficial mucosa; 540 nm penetrates deeper** → enhances **tumour vasculature**.
- **Detects more papillary tumours and CIS** (initially about 13% more tumours), but **false-positive rate about 60%**.
- **NBI-assisted TUR** is more thorough; a **randomised trial showed no overall recurrence difference**, but **reduced short-term (≤ 3 months) recurrence in low-risk tumours** — more thorough surface resection without altering high-risk recurrence patterns.

	Blue-light (HAL)	NBI
Principle	Porphyrin accumulation → red fluorescence	415/540 nm light absorbed by haemoglobin
Needs intravesical agent	Yes (catheter instillation)	No
Main gain	CIS detection	Papillary and CIS detection

	Blue-light (HAL)	NBI
Weakness	Cost; randomised trial showed no 3-year recurrence benefit	High false-positive rate

- **OCT and confocal laser endomicroscopy** are named as enhancement techniques; the chapter gives no further detail.
- **Enhanced techniques are included in AUA guidelines**, when available, to improve detection and completeness of TUR.

Complications of Transurethral Resection of Bladder Tumour and Bladder Biopsy

- **Common:** haematuria, infection, irritative symptoms. **Less common:** clot retention, TUR syndrome, perforation requiring cystorrhaphy, trigonal or urethral scarring.
- **Major complications (uncontrolled haematuria, perforation)** in 1–6.7%; subclinical extravasation on cystography in up to half.
- **Perforation risk factors:** elderly, large posterior wall tumours, multiple prior intravesical courses.
- **Prevention:** avoid overdistension; paralysis for lateral wall tumours (obturator reflex); **stage resection** of large or apparently invasive tumours.
- **Extraperitoneal (majority) vs intraperitoneal (dome):**

Extraperitoneal	Intraperitoneal
Majority of perforations	Dome resections
Prolonged urethral catheter drainage	Less likely to close → open or laparoscopic repair ; open repair allows washout to limit seeding

- **Tumour seeding after perforation: low risk**, but extravesical recurrences reported; higher apparent risk after surgical repair likely reflects **selection** of severe cases.
- **Perforation → more recurrence and worse disease-free survival, but not worse cancer-specific or overall survival** (more patients underwent cystectomy).
- **TUR syndrome** uncommon (especially bipolar); managed as in transurethral resection of the prostate (TURP).

KEY POINTS: ENDOSCOPIC SURGICAL MANAGEMENT

- **TURBT removes all visible tumour** and provides specimens for stage and grade.
- **Repeat resection within 6 weeks** for high-grade disease — **recommended for T1 and incomplete resection, considered for high-risk high-grade Ta** — especially if no muscle in the initial TURBT.
- **Sample all suspicious lesions; random biopsies are not required in low-risk patients.**
- **Office fulguration and observation** for selected low-risk patients.
- **Fluorescence cystoscopy and NBI improve visualisation** of inconspicuous tumours and may reduce recurrence after TUR.

Concept — *TURBT is a staging operation before it is a therapeutic one: completeness and muscle in the specimen are the two quality levers, and repeat resection, en bloc technique and enhanced imaging all exist to serve them.*

INTRAVESICAL THERAPY

Perioperative Intravesical Therapy

- **Rationale:** kill floating tumour cells **before implantation**; recommended by multiple guideline panels.
- **Meta-analysis:** single perioperative instillation **reduces recurrence in low-risk and in multiple tumours**; MMC, epirubicin and pirarubicin effective; thiotepa not (limited data, diluted doses).
- **MMC timing:** within **6 hours** reduces recurrence; at **24 hours** it does not — consistent with an anti-implantation mechanism. **Benefit persists even with ongoing adjuvant instillations** (randomised). **Level 1 evidence**, greatest for **initial solitary low-grade papillary tumour**.
- **Neoadjuvant MMC** (1 day and 4 hours before TURBT): small phase II trial, marked relative recurrence reduction, well tolerated.
- **Gemcitabine is now the standard perioperative agent in the United States** (efficacy, cost, safety; level 1 evidence).
- **SWOG S0337** (Southwest Oncology Group; landmark): **gemcitabine 2 g in 100 mL saline for 1 hour vs saline immediately post-TURBT** for suspected low-grade NMIBC; **4-year recurrence 35% vs 47%**; **low-grade only 34% vs 54%**; **34% relative reduction**; **no difference in adverse effects**.
- **An earlier negative gemcitabine trial** used **>20 hours of saline irrigation** in both arms → **continuous irrigation itself may reduce recurrence**; **sterile water and continuous saline irrigation** both show potential.
- **Complications:** irritative symptoms commonest; **serious sequelae and rare deaths, especially with perforation**; chemical cystitis, **cutaneous desquamation**, **contracture**, **calcified eschar**, harder later cystectomy.
- **Withhold chemotherapy with extensive resection or suspected perforation.**
- **BCG must never be given immediately after TUR** — risk of intravasation of live bacteria, sepsis and death.

KEY POINTS: PERIOPERATIVE INTRAVESICAL THERAPY TO PREVENT TUMOUR IMPLANTATION

- **Single-dose chemotherapy within 6 hours** reduces recurrence of low-risk tumours, most at **initial solitary low-grade papillary** presentation.
- **Incremental benefit in recurrent or multiple tumours is limited.**
- **Avoid with suspected perforation or extensive resection.**

Concept — *The single instillation is an anti-implantation measure, not a treatment of residual disease: hence the 6-hour window, the absolute bar on BCG at this point, and the largest benefit where implantation — not biology — drives recurrence.*

Intravesical Immunotherapy

Bacille Calmette-Guérin

- **BCG = attenuated mycobacterium** (tuberculosis vaccine) with antitumour activity. **Morales (1976)** introduced it with an added percutaneous dose, later dropped after intravesical success.
- **Administration sequence:** powder **reconstituted in 50 mL saline** → **start 2–4 weeks after TURBT** (re-epithelialisation) → **urinalysis before each instillation** (exclude infection or significant bleeding) → urethral catheter → **retain 1–2 hours** → **restrict fluid, diuretics and caffeine beforehand**.
- **Traumatic catheterisation** → **delay several days**. Active urinary tract infection (UTI) usually delays treatment; **asymptomatic bacteriuria may not require delay**.

- **Rotating the patient has no evidence; bleach-cleaning the toilet** is common advice though close-contact infection is undemonstrated.

BOX 130.1 CONTRAINDICATIONS TO BACILLE CALMETTE-GUÉRIN (BCG) THERAPY

Absolute	Relative	No or insufficient data
Immunosuppressed/ immunocompromised*	UTI (intravasation risk)	Prosthetic materials (no increased risk shown)
Immediately after TUR (intravasation, septic death)	Liver disease (precludes isoniazid if sepsis)	Ureteric reflux
Previous BCG sepsis	Previous tuberculosis (theoretical)	Anti-tumour necrosis factor drugs (theoretical sepsis risk)
Gross haematuria (intravasation)	Poor performance status	
Traumatic catheterisation (intravasation)	Advanced age	
Total incontinence (cannot retain)		

*Recent small series suggest immunosuppression may not be an absolute contraindication. From Ehlers S, J Rheumatol Suppl, 2005.

- **Unifying principle:** most absolute contraindications are **breaches of the urothelium → intravasation**.
- **Trap:** the box lists immunosuppression as absolute, but the text states **BCG can be used safely in transplant recipients and patients on steroids, disease-modifying agents or chemotherapy**, without excess sepsis — **response is merely weaker**, especially after transplant.

Mechanism of Action

- **Sequence:** BCG binds **fibronectin** in the bladder wall → local immune response: **cytokines in urine and wall** and influx of **granulocytes, mononuclear and dendritic cells** → cell-mediated cytotoxicity and an **antiangiogenic state**.
- **Cytokines:** tumour necrosis factor- α (TNF- α), granulocyte-macrophage colony-stimulating factor (GM-CSF), interferon- γ (IFN- γ), interleukins (IL) 1, 2, 5, 6, 8, 10, 12, 18. **Tumour necrosis factor-related apoptosis-inducing ligand (TRAIL)** release is key and associated with response.
- **Preferential IFN- γ , IL-2, IL-12 = T-helper type 1 (Th1) response** (effective); concomitant **IL-10 → suppressive Th2 response**.
- **Strains differ in immune pattern, but no strain is superior.**
- **Response limited by immunosuppressive disease and advanced age.**

Bacille Calmette-Guérin Treatment of Carcinoma In Situ and High-Risk Disease

- **BCG is the most used intravesical agent in the United States** (chemotherapy more used elsewhere); **TICE BCG is approved by the Food and Drug Administration (FDA)**.
- **Before BCG, CIS progressed about 7% per year. Initial tumour-free response up to 84%.**
- **Durability:** about **half have a durable response** (median 4 years); **only about 30% free of recurrence or progression at 10 years**, most events within 5 years → **close follow-up mandatory**. **Non-responders progress far more often than responders. Contemporary cohorts with adequate BCG show better outcomes** (high-grade recurrence-free survival at 12 months about 81–89%).
- **AUA:** BCG is preferred initial treatment for high-risk tumours (especially CIS); for intermediate risk, an option up front or **preferably after intravesical chemotherapy**.

- **BCG vs chemotherapy: BCG superior to MMC only when BCG includes maintenance** (vs 6-week MMC); **BCG reduces recurrence and progression; chemotherapy does not affect progression; higher complete response and durability with BCG; but more cystitis, dysuria, haematuria and fever.**

Bacille Calmette-Guérin Treatment of Residual Tumour

- **BCG eradicates residual papillary tumour in about 60% but must not replace resection; its primary use is adjuvant.**
- **CIS persisting after induction → second induction or first maintenance (3 weeks) yields complete response in a further 30–50%.**
- **Prostatic urethral mucosal/superficial duct carcinoma: BCG about 50% tumour-free; a limited TURP or fulguration reduces burden and improves BCG contact.**

Bacille Calmette-Guérin Therapy to Prevent Recurrence

- **6-week course after TURBT reduces recurrence (on average about 40%).**
- **Maintenance reduces recurrence more than induction alone (meta-analysis). 3 years of maintenance is supported; intermediate risk may be managed with 1 year.**
- **High-risk T1: BCG substantially lowers recurrence and progression vs TUR alone; multiplicity and associated CIS raise progression risk.**
- **T1a vs T1b substaging did not improve prediction of recurrence or progression after BCG.**

Impact of Bacille Calmette-Guérin on Progression

- **Progression is BCG's greatest value.** An early randomised trial showed **delayed progression** vs TUR and fewer cystectomies for CIS, with later cystectomy; yet **only about a quarter were alive with an intact bladder at 10–15 years.**
- **Meta-analyses: BCG reduces progression (vs chemotherapy and vs MMC specifically); benefit concentrated in trials using maintenance.**
- **No chemotherapy trial has significantly reduced progression.**
- **Long-term survival benefit of BCG is not fully defined.**

Determining Optimal Bacille Calmette-Guérin Treatment Schedule

- **Optimal dose and schedule not established; 6-week induction alone is insufficient — maintenance is requisite (AUA).**
- **Second induction: additional response about 25% (prophylaxis) and 30% (CIS). Beyond a second course, progression risk 20–50%; roughly 7% added progression risk per extra course.**
- **Response at 6 months predicts prognosis.**

LANDMARK BACILLE CALMETTE-GUÉRIN MAINTENANCE TRIALS

Trial	Population	Comparison	Result	Verdict
SWOG maintenance trial (Lamm, 2000)	Recurrent Ta, T1, CIS	Induction 6 weeks + 3 weekly instillations at 3 and 6 months, then every 6 months to 3 years vs induction alone	Median recurrence-free survival 76.8 vs 35.7 months ; 5-year survival 83% vs 78%; no toxicity above grade 3; only 16% completed full schedule, most dropouts in first 6 months	Maintenance superior despite poor completion; most common schedule used today

Trial	Population	Comparison	Result	Verdict
EORTC 30962	Intermediate and high risk Ta/T1	Full vs one-third dose; 1 vs 3 years maintenance (at 3, 6, 12 months; high risk then 6-monthly to 3 years)	Discontinuation for toxicity only about 7–8%	Intermediate: full dose, 1 year. High: full dose, 3 years
NIMBUS	High-grade NMIBC	Reduced frequency (9 doses: weeks 1, 2, 6; then 2 doses at 3, 6, 12 months) vs standard (15 doses, 1 year)	Recurrence doubled ; stopped at 12-month median follow-up	Reduced frequency inferior

- **Current standard:** responders with **high-risk disease** → **3 years full-dose maintenance**; **intermediate risk** → **1 year** (SWOG schedule, supported by EORTC 30962).
- **Trap:** reduced dose or schedule may suffice for **papillary-only** disease in theory, but **no equivalent regimen has been proven**; NIMBUS showed reducing frequency fails.

Antibiotics, Isoniazid and Response Prediction

- **Routine antibiotic prophylaxis during BCG may impair efficacy**; quinolones were thought to reduce BCG viability.
- **But short-term quinolone (ofloxacin or prulifloxacin) for 24–48 hours after instillation reduces cystitis side effects** (most from the fourth instillation), **reduces delays and discontinuations, without affecting recurrence or progression** — widely used routinely.
- **Isoniazid (INH)** single dose or 24 hours around instillation: **mixed results**.
- **Response prediction:** **CyPRIT nomogram** (urinary cytokine panel at the sixth induction instillation) and **Oncuria** (10-biomarker urine immunoassay before BCG) both predict response/non-response well; **both need validation**.

BOX 130.2 MANAGEMENT OF BACILLE CALMETTE-GUÉRIN (BCG) TOXICITY

Grade	Features	Assessment	Management
1: moderate, <48 h	Mild/moderate irritative symptoms, mild haematuria, fever <38.5°C	Consider urine culture	Anticholinergics, phenazopyridine, analgesics, non-steroidal anti-inflammatory drugs
2: severe and/or >48 h	Severe irritative symptoms, haematuria, or symptoms >48 h	Urine culture, chest radiograph, liver function tests	Reduce dose to one-half to one-third on resumption; treat culture; consider isoniazid pretreatment; isoniazid + rifampicin until resolution, with pyridoxine (vitamin B6); no monotherapy; watch rifampicin interactions (warfarin); monitor liver tests
3: serious (haemodynamic change, persistent high fever)	Allergic (joint pain, rash)	As grades 1–2	Isoniazid + rifampicin + pyridoxine

Grade	Features	Assessment	Management
	Solid organ (liver, lung, kidney)		Stop BCG; isoniazid and rifampicin; infectious disease input if persistent; add ethambutol; prednisone for inadequate response or septic shock — never without effective antimycobacterial cover

- **Traps:** BCG is almost uniformly resistant to pyrazinamide — no role; cycloserine causes severe psychiatric effects — avoid; asymptomatic granulomatous prostatitis after BCG may mimic prostate cancer and needs no treatment.

Interferon

- **Interferons (IFN)** = glycoproteins with antitumour actions: inhibit nucleotide synthesis, upregulate tumour antigens, antiangiogenic, cytokine release, T- and B-cell and natural killer cell activation. IFN- α most studied; active at ≥ 100 million units (optimal dose and schedule undetermined).
- **As a single agent:** more expensive and less effective than BCG or chemotherapy; CIS complete response 20–43% (dose-dependent), long-term efficacy <15%; limited activity against T1; occasionally effective after BCG failure (15–20%).
- **Combination:** additive with epirubicin or MMC; BCG + IFN- $\alpha 2b$ vs BCG alone (randomised, BCG-naïve) showed no difference. Main benefit: allows BCG dose reduction to one-third or lower without loss of efficacy.

Bacille Calmette-Guérin Shortages

- **Three shortages in a decade.** Connaught strain paused 2012, discontinued 2017; TICE is now the only FDA-approved strain, made in a single plant supplying 70 countries; demand exceeds supply.
- **SWOG S1602** (about 1000 patients, enrolment complete): TICE vs Tokyo-172 vs Tokyo-172 with intradermal priming — may bring a second approved strain and test priming.
- **Shortage strategies:** prioritise high-risk NMIBC → dose reduction, especially for maintenance → split vials (viability ≥ 72 hours) → intravesical chemotherapy for intermediate (single agent) and high risk (doublet) → early radical cystectomy for highest risk. Updates: AUA BCG shortage page.

KEY POINTS: IMMUNOTHERAPY

- BCG is more effective but more toxic than intravesical chemotherapy.
- Use BCG sparingly in low-risk disease and during shortages.
- Infectious complications are managed per Box 130.2.
- BCG reduces recurrence and is the only agent shown to delay or prevent high-grade progression.
- Maintenance improves outcomes (especially high risk) and should follow response to induction, as tolerated.
- BCG is contraindicated with disrupted urothelium (sepsis).

Concept — BCG works only as a sustained Th1 immune stimulus: that is why maintenance is what separates it from chemotherapy, why it is the only agent that alters progression, and why every contraindication is either a breached urothelium or a host who cannot mount the response.

Intravesical Chemotherapy

- **Single perioperative dose:** clear recurrence benefit. Adjuvant induction $\pm$ maintenance also reduces recurrence; MMC and gemcitabine most used.

- **BCG vs chemotherapy (adjuvant):** a SWOG trial found less progression with BCG than doxorubicin; BCG with maintenance lowers recurrence vs MMC, doxorubicin, epirubicin and thiotepa. Chemotherapy persists because of **no infectious risk and better tolerability**.
- **Intermediate risk: single-agent MMC or gemcitabine is an attractive alternative to BCG** with similar efficacy, especially for **low-grade multifocal recurrent disease**. Schedule: **6-week induction, then monthly maintenance for 1 year**; AUA: clinicians may use maintenance.

TABLE 130.5A AND 130.5B COMPARISONS AMONG INTRAVESICAL AGENTS

Agent	Class/ mechanism	Dose	Perioperative	Risk group	Cystitis (%)	Signature toxicity	Dropout (%)
Doxorubicin	Anthracycline: intercalation, topoisomerase II , free radicals	50 mg/ 50 mL	Yes	Low– intermediate	20–40	Fever, allergy, contracted bladder (5%) , cardiotoxicity, myelosuppression	2–16
Epirubicin	Anthracycline	50 mg/ 50 mL	Yes	Low– intermediate	10–30	Contracted bladder rare	3–6
Valrubicin	Anthracycline	800 mg/ 75 mL	No	Salvage	Mild	UTI, abdominal pain, asthenia	<10
Thiotepa	Alkylating: inhibits DNA/ RNA synthesis	60 mg/ 30 mL	Yes	Low– intermediate	10–30	Myelosuppression (8–19%) , rash, mucositis	2–11
Mitomycin C	Alkylating	40 mg/ 20 mL	Yes	Low– intermediate	30–40	Rash (8–19%) , severe skin reaction, contracted bladder 5%, encrustation cystitis	2–14
Gemcitabine	Nucleoside analogue: chain termination	2 g/ 100 mL	Yes	Low– intermediate; salvage	Mild	Occasional nausea; severe chemical cystitis 1–3%	<10
Docetaxel	Taxane: stabilises microtubules	75 mg/ 100 mL	—	—	Mild	Myelosuppression, hand-foot syndrome	—
Paclitaxel	Taxane	500 mg/ 100 mL	—	—	Mild	Myelosuppression, hand-foot syndrome	—
BCG	Innate and adaptive immunity	1 vial/ 50 mL	No	Intermediate– high	60–80	Fever 5%, flu-like 10%, severe cystitis 2–5% , granulomatous prostatitis 1%, BCG sepsis 0.5%	5–10

Agent	Class/ mechanism	Dose	Perioperative	Risk group	Cystitis (%)	Signature toxicity	Dropout (%)
Interferon	Immune activation; JAK- STAT	50 MU/ 50 mL	No	Salvage	<5	Flu-like 20%, fever 10%	Rare

Tables 130.5A and 130.5B merged. BCG, bacille Calmette-Guérin; MU, million units.

- **Purpose:** matches **mechanism** → **signature toxicity**: anthracyclines → contracture; thiotepa → marrow; MMC → skin; BCG → infection; IFN → flu-like.
- **Traps:** only BCG, IFN and valrubicin are not used perioperatively; gemcitabine is the only agent listed for both low-intermediate and salvage; IFN dose in the table (50 MU) is lower than the ≥100 MU the text cites for activity.

Doxorubicin and Analogues

- **Doxorubicin (Adriamycin)** — anthracycline antibiotic; **modest recurrence benefit over TUR, no progression benefit; chemical cystitis in up to half**; reduced capacity reported.
- **Epirubicin:** reduces recurrence vs TUR alone (**single immediate dose and 8-week courses**); oral *Lactobacillus casei* during instillations further reduced recurrence in a Japanese study; **BCG beats epirubicin on recurrence and progression, with more adverse events.**
- **Valrubicin:** doxorubicin analogue, **FDA-approved for BCG-refractory CIS in patients unwilling or unfit for cystectomy; complete response only about one-fifth** → little used.

Mitomycin C

- **Alkylating agent; historical agent of choice.** Induction **40 mg weekly × 6.**
- **Progression slightly higher than with BCG (meta-analysis); reduces recurrence but not progression;** fewer side effects. Latest meta-analysis: **no recurrence difference vs BCG overall, but BCG better where maintenance used; MMC higher progression risk.**
- **Optimising delivery (landmark: Au, 2001): eliminate residual urine, overnight fasting/fluid restriction, sodium bicarbonate (reduces degradation), concentrate to 40 mg in 20 mL.**
- **Device-assisted MMC:**

Technique	Principle	Key point
Microwave/radiofrequency hyperthermia	Local heating with MMC	Marked recurrence reduction; high 2-year recurrence-free rate in high-grade disease
Electromotive drug administration (EMDA)	Pulsed current (up to 20 mA, 30 minutes) via lower-abdominal electrodes; tissue delivery increased 4–7 fold; higher peak plasma MMC	Lower recurrence than passive MMC; BCG alternating with EMDA-MMC in T1 → lower recurrence and progression, better survival than BCG alone; vs BCG alone, a later randomised trial showed a non-significant difference and more toxicity
Chemohyperthermia (Synergo radiofrequency-induced thermochemotherapeutic effect, RITE)	Radiofrequency catheter heats bladder to 42°C	Up to threefold lower recurrence vs plain MMC; also potentiates BCG
Hyperthermic intravesical chemotherapy (HIVEC)	Recirculating MMC heated to 42°C (Combat Bladder Recirculation System)	Neoadjuvant: chemoablation in about 60%; adjuvant: low 2-year recurrence (intermediate risk); similar to EMDA at 12 months

Technique	Principle	Key point
Reverse thermal gel MMC (UGN-102)	Liquid when cold, gel at body temperature → sustained release	Primary chemoablation of low-grade intermediate-risk tumours: about two-thirds complete response at 3 months, mostly durable to 12 months; phase III under way ; same platform approved for low-grade upper tract UC

Gemcitabine

- Minimal systemic absorption; safe and effective adjuvant option.
- After BCG failure: vs BCG reinduction — **better 2-year disease-free survival, no difference in progression or cystectomy**; SWOG phase II after two failed BCG courses: 2-year disease-free survival 21%.
- Intermediate risk: gemcitabine ≈ one-third-dose BCG for recurrence and progression.
- Role: intermediate-risk disease and selected BCG-exposed patients.

Combination Therapy

- Chemotherapy combinations initially gave **more local toxicity for modest gain**.
- Chemotherapy + BCG: sequential MMC-BCG showed **no advantage**; BCG monotherapy even **outperformed** a combination arm. **No benefit from sequential chemotherapy-BCG regimens except with electromotive MMC**. (An EORTC marker-lesion study gave about half complete response with sequential MMC and BCG.)
- Gemcitabine-docetaxel (Gem/Doce) — renewed interest (BCG-unresponsive disease, shortage); **most studied as primary adjuvant therapy for BCG-naïve high-grade NMIBC**.
- Regimen sequence: gemcitabine 1 g in 50 mL (sterile water or saline), 60–90 minutes → drain → docetaxel 37.5 mg in 50 mL saline, 60–120 minutes → weekly × 6 induction → monthly maintenance for 2 years in responders.
- Tolerability aids: oral sodium bicarbonate 1300 mg the night before and morning of treatment, ondansetron, anti-inflammatories.
- BCG-naïve outcomes: **high 2-year recurrence-free survival** in retrospective and small prospective series. **Phase III Gem/Doce vs BCG (induction + maintenance) is under way**.

KEY POINTS: INTRAVESICAL CHEMOTHERAPY

- **Clear effect on recurrence** (immediate post-TURBT and adjuvant); **no clear evidence on progression**.
- Side effects less severe than BCG and confined to the bladder.
- Combination chemotherapy has a role in BCG-refractory disease and an **emerging role in BCG-naïve high-risk disease**.

Concept — Chemotherapy is a **recurrence tool** and BCG a **progression tool**; device-assisted delivery and doublets are attempts to push chemotherapy across that line, and only doublets are now serious contenders.

DISEASE RECURRENCE AFTER BACILLE CALMETTE-GUÉRIN

- "BCG failure" has been poorly defined. Definitions have been **refined for trial design and to reflect biology** (FDA workshop, international consensus).

DEFINITIONS OF DISEASE AFTER BACILLE CALMETTE-GUÉRIN

Category	Definition	Implication
BCG-refractory (historical)	Persistent disease despite two induction courses, or induction + maintenance (usually within 6 months), or intolerance	Very high risk → strongly consider immediate cystectomy if young and fit
BCG-unresponsive	(1) Recurrent or persistent CIS or HG Ta/T1 despite adequate BCG, or (2) initial response then HG recurrence within 6 months (Ta/T1) or 12 months (CIS) of the last dose	Unlikely to benefit from more BCG → strongly consider radical cystectomy; trials/salvage if unfit
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction (HG Ta or CIS); for T1: ≥5 of 6 induction alone	Threshold for "unresponsive"
BCG-intolerant	Persistent disease due to inability to receive adequate BCG (toxicity)	Classified separately — not unresponsive
BCG-exposed	HG recurrence after BCG not meeting unresponsive criteria	Some benefit from further BCG
BCG-resistant	Persistent/recurrent CIS or HG Ta at the 3-month cystoscopy after induction	Reinduction or continue maintenance (after resection) → complete response 30–50%; often prematurely labelled unresponsive
Delayed relapse after adequate BCG	HG recurrence outside the window (>6 months Ta/T1, >12 months CIS); formerly intermediate (12–24 months) or late (>24 months) relapse	Later relapse → may respond to BCG rechallenge
Delayed relapse after inadequate BCG	Induction only / incomplete exposure (worsened by shortages)	Heterogeneous

- **Trap: HG T1 at the 3-month check after induction alone already qualifies as unresponsive; CIS or HG Ta at 3 months does not** (BCG-resistant → more BCG).

Assessment of Response

- **Biopsy after BCG:** necessity unclear, but **strongly consider in high-risk CIS at 6 months. Routine biopsy adds little if cystoscopy and cytology are negative; erythema + positive cytology** is the combination that yields positive biopsies. **Urine cytology is useful; UroVysion fluorescence in situ hybridisation (FISH) conversion to negative correlates with response.**
- **Extravesical vigilance:** intermediate/high-risk patients with recurrence, persistence or positive cytology after adequate therapy may need **prostatic urethral biopsy and/or upper tract evaluation. Prostatic urethral recurrence 24–30%; upper tract 0.8–25%. More bladder recurrences → more upper tract tumours. BCG-unresponsive patients: urethral or upper tract recurrence in half on long-term follow-up.**

Management of Recurrent High-Grade Disease

- **If initial treatment was chemotherapy → give BCG** (repeat chemotherapy courses fare poorly).
- **BCG-intolerant → salvage chemotherapy**, but high failure and progression.
- **Timely radical cystectomy for BCG-unresponsive HG NMIBC (EAU and AUA). Recurrent HG T1 after induction BCG has poor prognosis without cystectomy**, with high progression on further intravesical therapy.

Bladder-Preserving Options for Bacille Calmette-Guérin–Unresponsive Disease**TABLE 130.6 SALVAGE THERAPIES FOR BCG-UNRESPONSIVE DISEASE**

Outcome	Pembrolizumab*	Nadofaragene firadenovec*	Gemcitabine	Gem/MMC	Gem/Doce	N-803 + BCG†
3-month CR, CIS	41%	53%	47%‡	68%‡ (any time)	NR	71% (any time)
12-month CR, CIS	19%	24%	28%‡	48%‡ (2 yr 38%)	60% (2 yr 43%)	45%
12-month RFS, papillary	44%	44%	NR‡	NR‡	62% (2 yr 51%)	55%
Duration of CR (responders)	24.1 mo	9.7 mo	NR	26 mo	13.9 mo	19.2 mo
Schedule	IV q3 weeks × 2 years	Intravesical every 3 months	Weekly × 6, monthly × 1 yr	Weekly × 6, monthly × 1 yr	Weekly × 6, monthly × 2 yr	Weekly × 6, maintenance × 3
Grade 3–4 adverse events	13%	4%	5%	8%	3.3%	11%
Cystectomy-free	63%	71%	74%	79%	84.4%	87.5%

*FDA-approved. †Under FDA review at time of writing. ‡CIS and papillary patients grouped. CR, complete response; RFS, recurrence-free survival; NR, not reported.

- **Purpose:** benchmarks every bladder-sparing option against the same endpoints. **Pattern:** early CR of about 40–70% **decays by half or more by 12 months** — none rivals cystectomy for durability.
- **Discrepancies with the text:** pembrolizumab **duration of CR 24.1 months (table) vs about 16 months (text)** and papillary **12-month RFS 44% (table) vs 43.5% (text)** reflect **different data cuts** (updated vs original report); Gem/Doce **papillary RFS 62%/51% (table) vs 65%/52% (text)** — **table gives the papillary subgroup, text the whole cohort.**
- **Trap:** pembrolizumab has the highest grade 3–4 toxicity (immune-related); Gem/Doce the highest cystectomy-free rate among approved/standard options — **but no level 1 evidence.**

Sequential Intravesical Chemotherapy

- **Gemcitabine is the most active single agent** (2-year recurrence-free survival about one-fifth in mixed BCG-exposed/unresponsive patients).
- **Gem/MMC:** gemcitabine 1000 mg, 60–90 minutes → drain → MMC 40 mg, 60–90 minutes; durable recurrence-free rate about 38%.
- **Gem/Doce (more common):** multi-institutional data — **1- and 2-year high-grade recurrence-free survival about 65% and 52%, progression to muscle invasion under 4%, only about 3% could not complete, 84% avoided cystectomy; CIS complete response 60% at 1 year, 43% at 2 years. Needs confirmation; no level 1 or prospective evidence.**

Nadofaragene Firadenovec

- **Replication-incompetent adenovirus type 5** carrying the **IFN-α2b gene**, with **Syn3 polyamide surfactant** to enhance urothelial transduction; **FDA-approved** for BCG-unresponsive disease.
- **Pivotal single-arm trial:** 75 mL (3×10^{10} viral particles/mL) instilled for 1 hour every 3 months; **CIS complete response 53% at 3 months and 24% at 12 months; papillary high-grade RFS 44% at 12 months; progression to muscle invasion 3%; 71% cystectomy-free; serious adverse events 2%.**

- **Moderate response, favourable toxicity and convenient schedule.**

Checkpoint Inhibitors

- **Systemic immunotherapy blocking programmed death-1 (PD-1)/programmed death-ligand 1 (PD-L1) interaction**, which normally inhibits T-cell activation.
- **KEYNOTE-057 (landmark): pembrolizumab 200 mg IV every 3 weeks for up to 2 years; first systemic therapy FDA-approved for NMIBC (BCG-unresponsive CIS). CIS: 41% complete response at 3 months; 19% at 12 months; 63% avoided cystectomy. Grade 3–4 toxicity 13% (8% serious), no treatment-related deaths. Papillary cohort: 12-month RFS about 44%.**
- **Mild response; vigilance for immune-related adverse events.**

Photodynamic Therapy

- **Photosensitiser (systemic porfimer sodium [Photofrin] or intravesical HAL) → intravesical light irradiation → reactive singlet oxygen and free radicals → cytotoxicity.**
- **Good response in CIS and papillary disease, but bladder contracture/irritability (about half) and skin photosensitivity limited use. HAL with incoherent white light is better tolerated but yields few durable responses.**

Novel Agents

Agent	Class	Delivery	Key point
TAR-200	Continuous low-dose gemcitabine delivery system (pretzel-shaped)	Placed cystoscopically; exchanged every 3 weeks to week 24, then every 3 months × 6	Uniform drug contact, minimal absorption; very high early CR alone in small numbers; cetrelimab alone weaker; well tolerated
Cretostimogene (CG0070)	Replication-competent oncolytic adenovirus; E2F promoter (tumour-selective replication) + GM-CSF transgene	Intravesical, two 6-week inductions + maintenance; combined with pembrolizumab	About two-thirds CR at 12 months with pembrolizumab; phase III under way
N-803 (ALT-803)	IL-15 superagonist → CD8+ T-cell and natural killer cell activation	Intravesical with BCG	CIS CR 45% at 12 months; papillary about 55% disease-free; 84% avoided cystectomy; grade 3–4 events 4%

- **Nadofaragene vs cretostimogene: replication-incompetent (gene delivery of IFN- α 2b) vs replication-competent (oncolysis + GM-CSF).**

Radiation Therapy

- **Restricted to patients who refuse or are unfit for cystectomy after intravesical failure. TUR + radiotherapy: complete response 50–75%, but added benefit over TUR unclear; no effect on CIS; up to half progress → largely palliative.** Chemoradiation promising but not widespread; **role unclear** given new agents.

KEY POINTS: MANAGEMENT OF HIGH-GRADE RECURRENT DISEASE

- **Recurrence after initial intravesical therapy = high risk of recurrence or progression.**
- **After primary chemotherapy, give BCG.**

- **BCG-unresponsive** = recurrent/persistent CIS or HG Ta/T1 after adequate BCG, or **HG recurrence within 6 months (Ta/T1) or 12 months (CIS)** of the last dose → **strongly consider radical cystectomy**.
- **Unfit or declining cystectomy:** doublet chemotherapy, nadofaragene firadenovec, pembrolizumab, or trials.

Role of "Early" Cystectomy

- **Rationale:** BCG failure in CIS → about 50% progression; 3-month T1 failure after BCG → 82% progression; up to 20% of CIS patients die of UC within 10 years; each T1 recurrence carries 5–10% metastatic risk; residual tumour at re-resection → high rate of muscle invasion.
- **Aggressive intravesical therapy long term:** only about a quarter alive with intact bladder at 15 years; about half progress; one-third die of disease.
- **Immediate cystectomy for clinical T1:** accurate staging and **excellent 10-year disease-free survival (92%)**, falling to 64% if understaged muscle-invasive disease is found; up to 15% already have micrometastases; delay of even 12 weeks worsens survival.
- **"Timely" cystectomy** (Chang and Cookson) balances progression risk against morbidity; **10-year survival after cystectomy for NMIBC 67–92%**.
- **Trap:** patients who progress to muscle invasion under surveillance fare worse than those presenting with muscle-invasive disease.
- **Sexual- and nerve-sparing cystectomy** in this lower-risk group: **oncological safety unknown**. Neobladder availability reduces treatment delay.
- **Also consider cystectomy when tumour cannot be controlled endoscopically:** bulky, inaccessible (large bladder, urethral stricture).
- **Strongly consider radical cystectomy for:** high grade with deep lamina propria invasion, LVI, diffuse CIS, diverticular tumours, substantial distal ureteric or prostatic urethral involvement, BCG-unresponsive, too large or inaccessible, or patient preference for definitive therapy.
- **AUA:** radical cystectomy for fit patients with **persistent HG T1 on repeat resection**, or **T1 with CIS, LVI or variant histology**; and for **high-risk BCG-unresponsive disease**.
- **Partial cystectomy:** limited data; option in selected patients.

KEY POINTS: EARLY CYSTECTOMY

- **Radical cystectomy** is definitive with favourable outcomes; consider for:
- **BCG-unresponsive high-risk NMIBC;**
- **High progression risk:** variant histology, distal ureter and/or prostatic urethral involvement, LVI, deep lamina propria invasion, **persistent HG T1 at re-resection;**
- **Disease not amenable to complete resection** (size, volume, access).

Concept — The case for early cystectomy rests on an **asymmetry**: clinical T1 is systematically understaged, and disease that progresses under surveillance does worse than de novo muscle invasion — so the cost of waiting falls on exactly the patients who cannot afford it.

SURVEILLANCE AND PREVENTION

- Among the most expensive cancers per patient, largely from long-term surveillance.

- **Surveillance: cystoscopy + urine cytology**, the accepted **gold standard** despite subjectivity. **Adherence to risk-stratified schedules is a quality metric**, yet **fewer than half** follow a standard protocol.
- **Tools: AUA/SUO stratification (Table 130.4), EAU stratification, EORTC tables. Tailor to the individual.**

Cystoscopic Surveillance

- **Office flexible cystoscopy:** rapid, relatively painless; **papillary tumours readily seen**; CIS classically a **velvety red patch** (unreliable). **High positive predictive value**, but **appearance cannot reliably predict stage or grade**; sessile morphology and/or necrosis suggests high-grade, likely invasive disease.
- **Office blue-light flexible cystoscopy** detects more recurrences and CIS; requires **pre-procedure catheterisation and instillation of hexaminolevulinate (Cysview)**.
- **Comfort:** intraurethral anaesthetic gel is near-universal but **evidence conflicting**; **no benefit in women** (straighter urethra); **men watching the monitor report about half the pain**.
- **Technique:** **empty bladder beforehand**; **60 mL syringe aspiration** via irrigant port to clear clouding; **systematic survey mandatory**. **Air cystoscopy** aids views in active haematuria or where sterile fluid is unavailable.
- **Most predictive for recurrence:** tumour at the 3-month cystoscopy and number of tumours at initial resection.
- **Low-grade Ta clear at 3 months → annual cystoscopy is safe from 12 months.**
- **Duration:** AUA — **beyond 10 years case-by-case**; earlier work suggested **stopping at ≥5 years for low risk**; late high-grade recurrences temper enthusiasm for stopping.

TABLE 130.7 2016 (UPDATED 2020) AMERICAN UROLOGICAL ASSOCIATION AND SOCIETY OF UROLOGIC ONCOLOGY (AUA/SUO) GUIDELINE-SUGGESTED SURVEILLANCE STRATEGIES

Risk	Tumour status	Cystoscopy	Cytology/ markers	Upper tract imaging
Low	Solitary LG Ta	3 months, then annually from 12 months; consider stopping at ≥5 years	Consider	Not necessary unless haematuria
Intermediate	Multiple LG Ta; large tumour; recurrence at 3 months	Every 3–6 months for 1–2 years, 6–12 months for next 2 years, then annually; restart clock with each recurrence	Consider	Consider every 1–2 years, especially with recurrence; image for haematuria
High	Any HG (including CIS)	Every 3–4 months for 2 years, 6-monthly for 2 years, annually for life; restart clock with each recurrence	Cytology at same schedule; consider markers	Annually for 2 years, then consider lengthening

Modified from Chang SS et al., J Urol 2016 (updated 2022).

- **Pattern:** intensity front-loaded in years 1–2; **only high risk is lifelong**; **cytology is mandatory only for high risk**.
- **Trap:** the table's "**consider cessation at ≥5 years**" for low risk sits alongside the text's "**beyond 10 years case-by-case**" — the former is the table's low-risk option, the latter the general guideline statement.

Urine Cytology

- **No urine test has sufficient accuracy to replace cystoscopy. Cytology = most studied**; microscopic evaluation of shed cells (Papanicolaou); **subjective**, needs adequate volume and cellularity.
- **The Paris System** standardised reporting to reduce inconclusive results.

- **High specificity:** a positive result implies malignancy **regardless of cystoscopy or imaging**; **poor cohesion of high-grade cells, especially CIS, raises yield.**
- **Persistently positive cytology with negative cystoscopy and upper tract imaging → substantial risk of genitourinary cancer within 2 years → further workup warranted.**
- **Sensitivity is low — even for high grade in some series. Cytology = high specificity, low sensitivity → an adjunct.**

Tumour Markers

- **Most markers: adequate sensitivity, poor specificity → false positives and extra testing. Cell-based or protein-based.** Adoption limited by **lack of prospective data** and study heterogeneity; one study linked marker use with better survival.

Marker	Type/target	Strength	Weakness
BTA stat / BTA TRAK	Complement factor H-related protein (point-of-care / quantitative)	More sensitive than cytology, especially low grade	False positives with inflammation, infection, haematuria; lower specificity
uCyt/ ImmunoCyt	Hybrid cytology + 3 fluorescent monoclonal antibodies	Not affected by haematuria or infection; high negative predictive value (NPV) for atypical cytology	Complex, operator-dependent; needs cellularity; FDA-approved for follow-up only
Nuclear matrix protein 22 (NMP22) / BladderChek	Nuclear mitotic apparatus protein released at apoptosis (ELISA / point-of-care)	More sensitive than cytology; adds to cystoscopy sensitivity	False positives: stones, infection, inflammation, haematuria, instrumentation; less specific than cytology
UroVysion FISH	Aneuploidy of chromosomes 3, 7, 17 + 9p21 deletion	Highest specificity of markers; detects CIS far better than cytology ; "anticipatory positive" precedes visible tumour (3–15 months); negative predicts no recurrence within a year; not affected by haematuria/ inflammation; marker of BCG response; clarifies atypical cytology	Nondiagnostic reports behave as negative
CxBladder	Gene expression (CxTriage, CxDetect, CxMonitor)	High sensitivity and NPV → may reduce surveillance cystoscopies	—
Others	Xpert Bladder, ADXBLADDER, Oncuria, Urodiag, VisioCyt	Promising	Need further study

FIGURE 130.9 — UROVYSION FISH

- **Abnormal enlarged cell with three copies each of chromosomes 3 (red), 7 (green) and 17 (blue) and homozygous 9p21 deletion (yellow).**
- **Marker choice depends on the endpoint:** to select patients for operating-room biopsy → **high specificity**; to lengthen surveillance intervals → **high sensitivity** (especially for high grade). **Accuracy rises with stage and grade.**
- **AUA: do not use markers to replace cytology in surveillance; consider them to assess response to intravesical therapy and to adjudicate atypical or equivocal cytology.**

- **Upper tract markers: little data. Selective upper tract cytology** increases yield but is **falsely positive** in the presence of a bladder tumour → not recommended for most.

Extravesical Surveillance

Upper Tract

- **Upper tract UC after NMIBC is uncommon** (registry about 0.8%), rising with risk — low under 1%, high risk about 10%, and higher still in very high-risk or BCG-treated populations (13–18%). Surveillance is worthwhile mainly with haematuria or high-grade tumour, especially near the ureteric orifice. Yield of routine upper tract imaging is very low.
- **AUA: no routine upper tract imaging in asymptomatic low-risk patients; consider every 1–2 years for intermediate and high risk** (little supporting data).
- **Imaging: multiphasic CT urography preferred; magnetic resonance urography and renal ultrasonography** in select patients; **retrograde pyelography** for contrast allergy or renal insufficiency, often at TURBT.
- **Upper tract disease, though rare, carries 40–70% mortality.** In BCG-treated high-risk disease, risk is greatest in the first 5 years (median detection about 5 years) but **persists ≥15 years**.

Prostatic Urethra

- **Secondary prostatic urethral/ductal involvement in high-risk NMIBC: 10–15% at 5 years; 20–40% at 10 years.**
- **Refractory disease → prostatic fossa recurrence in about one-third, nearly half of these fatal.**
- **Any high-grade prostatic urethral involvement = high risk.** Risk factors: CIS, multifocality, bladder neck/trigone tumours.
- **Ductal high-grade involvement → radical cystoprostatectomy; consider urethrectomy**, especially with tumour at or near the margin.
- **Individualise surveillance to bladder and extravesical risk.**

Prevention

- **Smoking cessation counselling is key**, though its impact on recurrence is uncertain; **increased fluids and a low-fat diet** are recommended. (Stated in the key points only; no body text in the chapter.)

KEY POINTS: SURVEILLANCE AND PREVENTION

- **Cystoscopic surveillance is required**, individualised to the **risk of the most recently resected tumour**.
- **Table 130.7** gives reasonable protocols.
- **Markers improve on cytology's sensitivity, but most are less specific.**
- **Smoking cessation counselling is key**; impact on recurrence uncertain.
- **Increased fluids, smoking cessation and a low-fat diet** are recommended.

Concept — *Surveillance intensity mirrors the recurrence–progression split: low risk needs to be spared tests, high risk needs lifelong vigilance extending beyond the bladder, and every marker is judged by whether the question is "who needs a biopsy" (specificity) or "who can wait" (sensitivity).*

PATIENT-FOCUSED OUTCOMES AND ADVOCACY

- **Burden:** lifelong endoscopy and repeated instillations → **reduced quality of life and financial toxicity**, plus anxiety about recurrence and progression.
- **CISTO trial** (Comparison of Intravesical Therapy and Surgery as Treatment Options): large **pragmatic observational** study of **recurrent high-grade NMIBC after BCG failure** — quality of life, survival, progression, preferences and caregiver burden, using validated patient-reported outcomes.
- **UGN-102** (sustained-release MMC reverse thermal gel): framed as **more patient-centred**, avoiding the urinary, sexual, physical and mental-health decline of repeated TURBT.
- **Bladder Cancer Advocacy Network (BCAN):** patient and caregiver support, education, research sponsorship, advocacy.